



ELE1204: Digital Electronics and Logic Design

Lecture 06 (BSCS)

Registers

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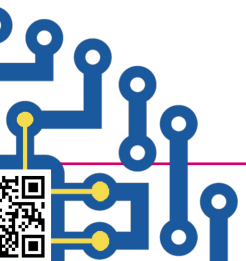


Lecture Objectives and Learning outcomes

The Objectives of this lecture are :

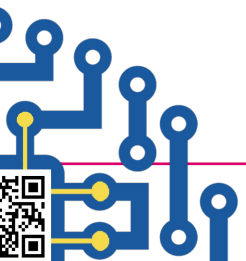
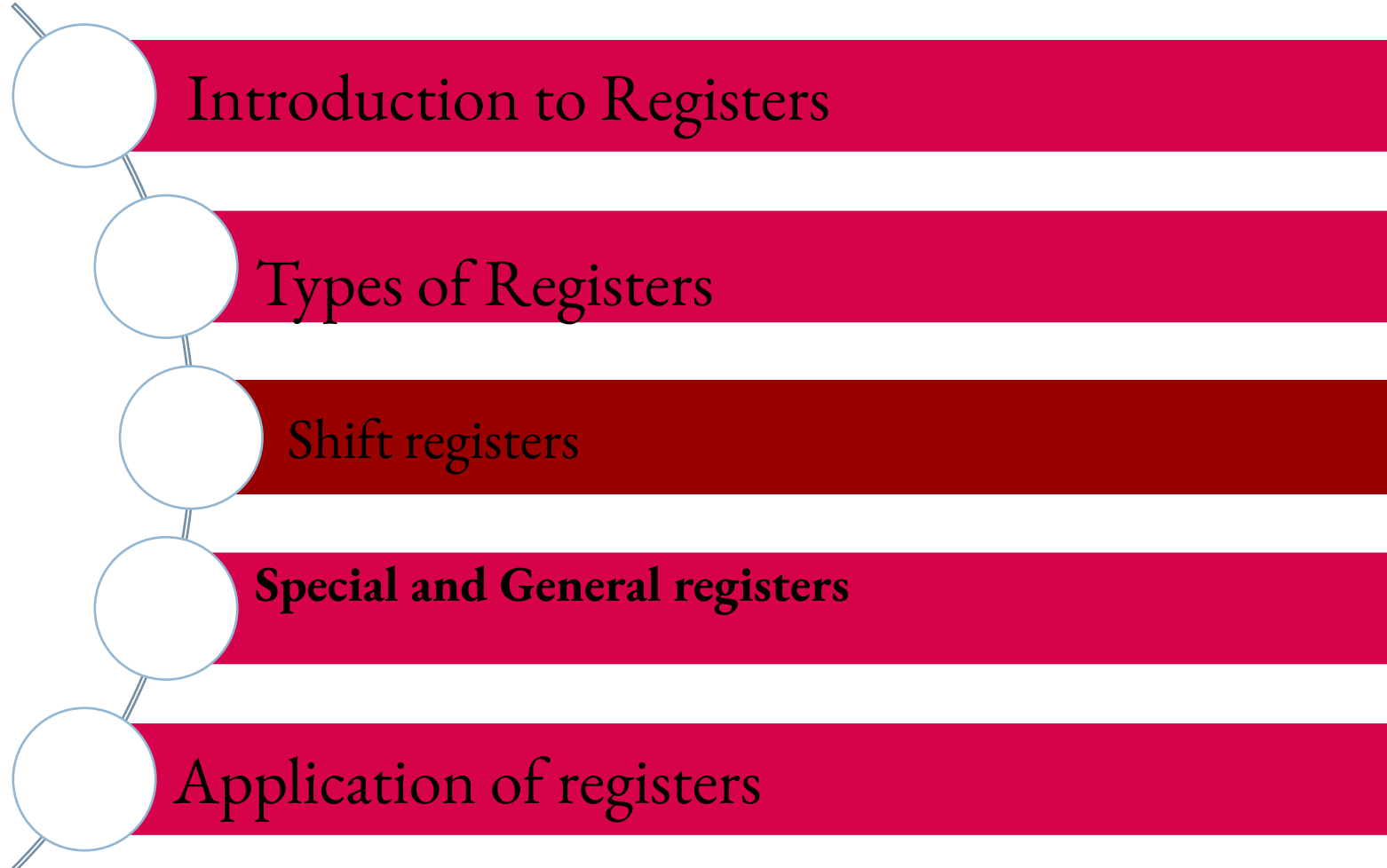
- ☐ Explain how registers store and manipulate data within a computer processor.
- ☐ Differentiate between General-Purpose Registers (GPRs) and Special-Purpose Registers (SPRs).
- ☐ To understand the different types of shift registers

NOTE:: SMART = **S**pecific, **M**easurable, **A**chievable, **R**elevant, **T**imebound





Lecture Overview





Introduction to Registers

Definition of Registers

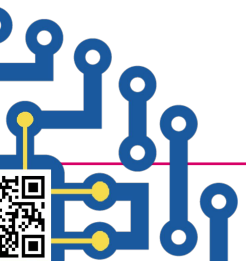
A **register** is a **group of flip-flops** used to **store multiple bits of data**. It is used for **temporary storage, data transfer, and manipulation** in computer processors and digital circuits.

A register is a group of flip-flops, each one of which shares a common clock and is capable of storing one bit of information

A circuit with flip-flops is considered a sequential circuit even in the absence of combinational gates

<https://www.youtube.com/watch?v=-paFaxtTCkI>

4bit register needs 4 flipflops



Shift Registers

A register capable of shifting the binary information held in each cell to its neighboring cell, in a selected direction.

A **shift register** is a sequential logic circuit that **stores and moves data (bits) in a specific direction** with each clock pulse.

D FlipFlops are used based on their simplicity.

<https://www.youtube.com/watch?v=unorn9n-UpE>

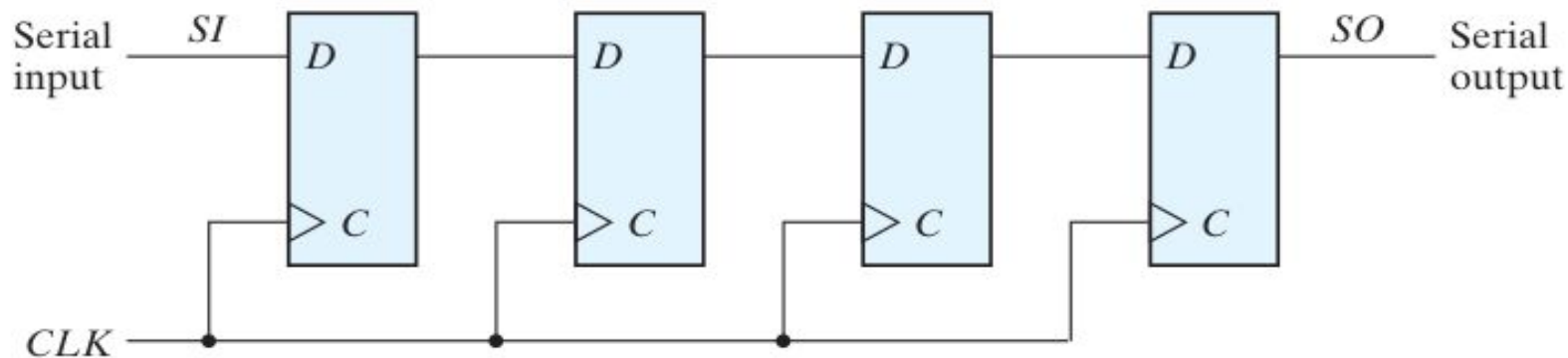
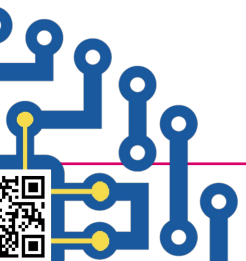


FIGURE 6.3
Four-bit shift register



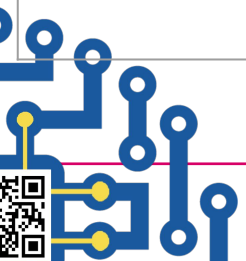


Types of shift Registers



Registers are classified based on **how data is stored and transferred**:

Type of Register	Function	Example
Serial-In Serial-Out (SISO)	Data is shifted in and out one bit at a time.	Data transmission in communication systems.
Serial-In Parallel-Out (SIPO)	Data is shifted in serially but read in parallel .	Used in ADC (Analog-to-Digital Converters).
Parallel-In Serial-Out (PISO)	Data is loaded in parallel but shifted out serially .	Used in data transmission circuits.
Parallel-In Parallel-Out (PIPO)	Data is loaded and retrieved in parallel .	Used in microprocessors for temporary storage.



Types of shift Registers

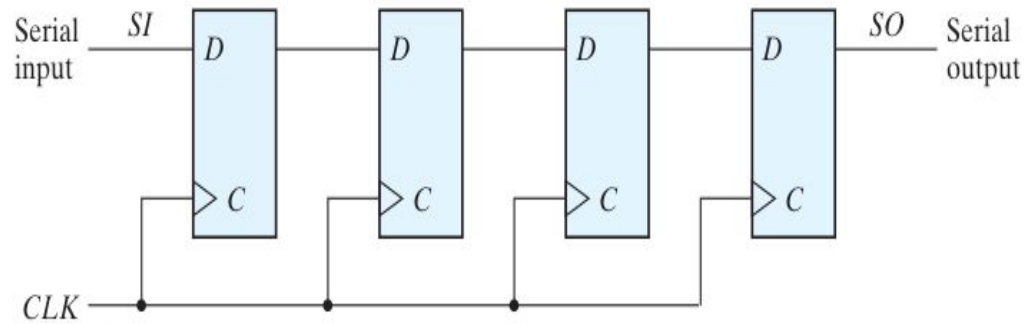
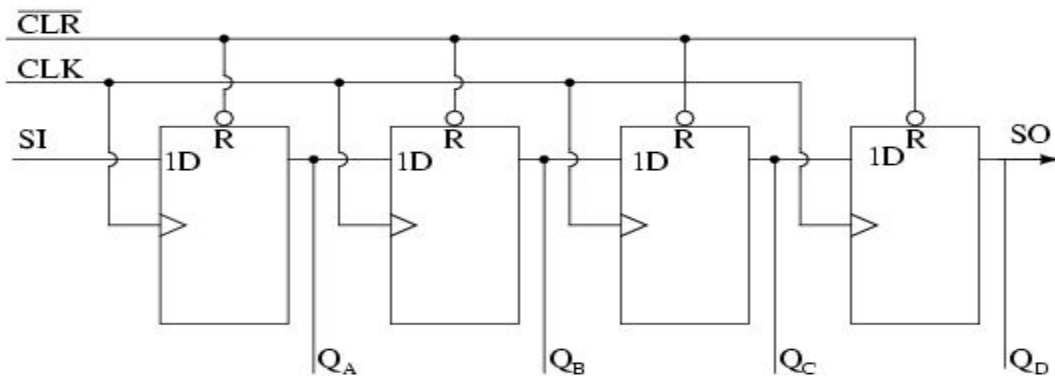
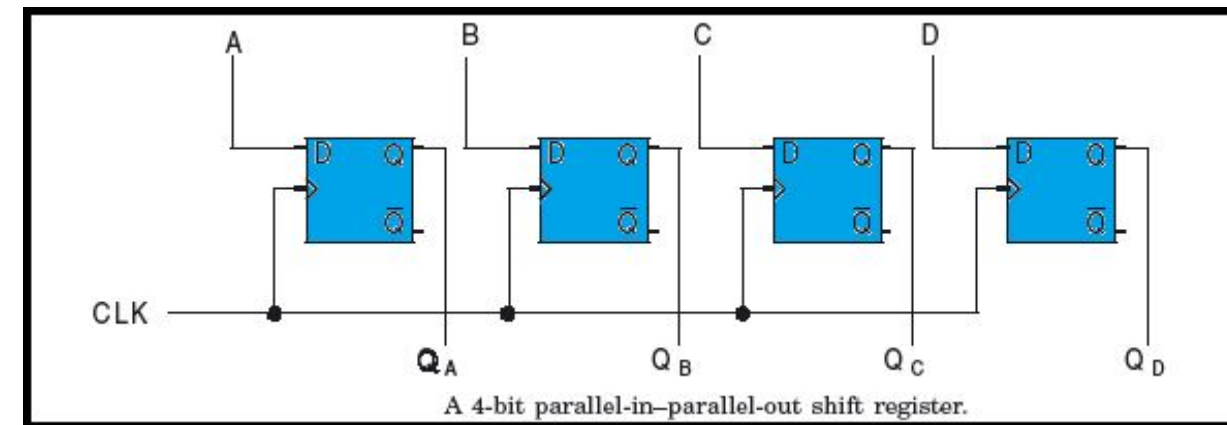
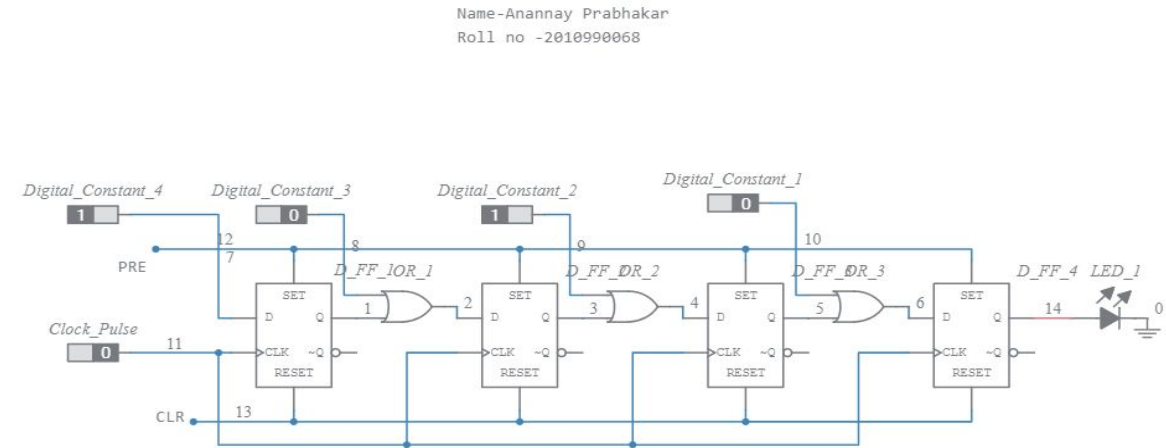


FIGURE 6.3
Four-bit shift register



Serial-in/ Parallel out shift register details



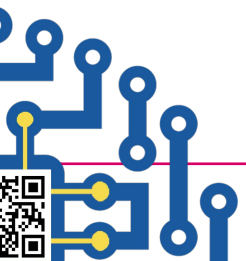


Types of Registers (Computer architecture)

General-Purpose Registers (GPR)

Registers can be classified into different types based on their **function and purpose** in a computer system.

- **Function:** Store intermediate values during calculations or data processing.
- **Example:** In x86 processors, registers like **AX, BX, CX, and DX** are general-purpose.
- **Usage:**
 - a. Temporary storage for arithmetic/logical operations.
 - b. Holding variables in programs.
 - c. Data transfer between CPU and memory.





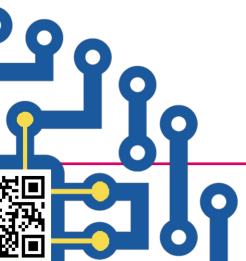
Types of Registers

General-Purpose Registers (GPR)

Accumulator Register

This is a General-Purpose Register (GPR) used in the Arithmetic Logic Unit (ALU) to store intermediate results of arithmetic and logic operations.

- ♦ The ALU performs operations like addition, subtraction, AND, OR, and shifting, and the result is stored in the Accumulator before being transferred to memory or another register.
- ♦ In early microprocessors, the Accumulator was the main register used for computations.

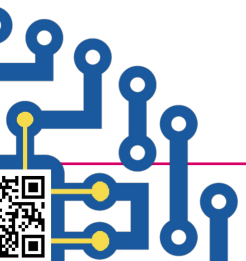




Types of Special purpose Registers

Instruction Register (IR)

- **Function:** Holds the current instruction being executed by the CPU.
- Role in the CPU:
 - Stores the binary machine code of the instruction fetched from memory.
 - Sends the instruction to the Control Unit (CU) for decoding and execution.
 - Helps in coordinating the execution of instructions in sequential order.

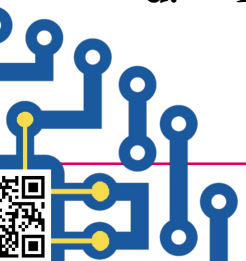




Types of Registers

Memory Address Register (MAR)

- **Function:** Holds the **address** of the memory location that is currently being accessed (read/write operation).
- **Role in the CPU:**
 - a. Fetches instructions: Stores the memory address of the next instruction to be retrieved.
 - b. Reads/Writes data: Works with MDR to fetch/store data at a specific memory address.
 - c. Helps in direct memory access (DMA) operations.
- **Example Use Case:**
 - a. The CPU needs to fetch instruction 1010 from memory.
 - b. The MAR holds the address 1010.
 - c. The MAR sends the address to memory via the address bus.
 - d. The instruction/data at memory location 1010 is retrieved and placed in the MDR.





Types of Special Registers

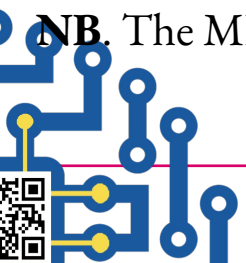
Memory Data Register (MDR)

- **Function:** Temporarily stores **data or instructions** being transferred to/from memory.
- **Role in the CPU:**
 - a. Acts as a **buffer** between the CPU and main memory.
 - b. Holds **data read from memory** before sending it to the CPU.
 - c. Stores **data to be written to memory** before transferring it to the address specified by the MAR.

✓ Example Use Case:

1. The **CPU wants to read data from memory address 1100.**
2. The **MAR sends 1100 to memory.**
3. The data stored at **address 1100** is placed into the **MDR.**
4. The **CPU retrieves the data from the MDR** for processing.

NB. The MDR is like a **clipboard that temporarily holds data before it is used.**

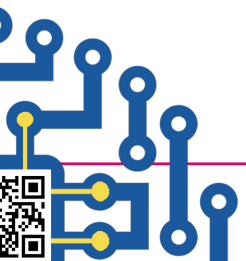




Registers

Application of Registers

- Temporary storage in microprocessors (e.g., **Instruction Register (IR)**, **Accumulator Register (AC)**).
- Data transfer in communication systems.
- Buffer storage in I/O devices.





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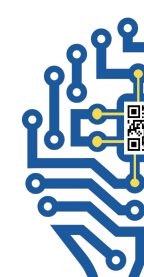
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